

CYCLeSS Demo Notebook: What It Shows

A walkthrough of the data loading and yield regression demo, for presenting to Evie and the team.

The premise

The notebook demonstrates two things Scivision-style tooling expects: loading data through a standard, discoverable mechanism, and running an analysis with an off-the-shelf tool. One caveat worth stating up front: CYCLeSS itself was never submitted to the Scivision catalog. The catalog is scoped to computer-vision tasks, and this data is tabular, so this shows the loading pattern Scivision would use, not an actual catalog listing.

Loading the data

First, `data.yml` is opened through intake, the same catalog-config format `scivision.load_dataset()` consumes. This registers three sources: `cycless_yield_satellite`, `cycless_soil`, `cycless_climate`. The actual reading is then done with plain pandas against those same folders, since CYCLeSS's per-band, per-year file layout doesn't map cleanly onto intake's generic CSV driver. The catalog serves as the documented, standardised description of where the data lives, while pandas does the real work.

Building the modelling table

Three pieces get loaded and joined into one 936-row table with zero nulls after dropping incomplete rows:

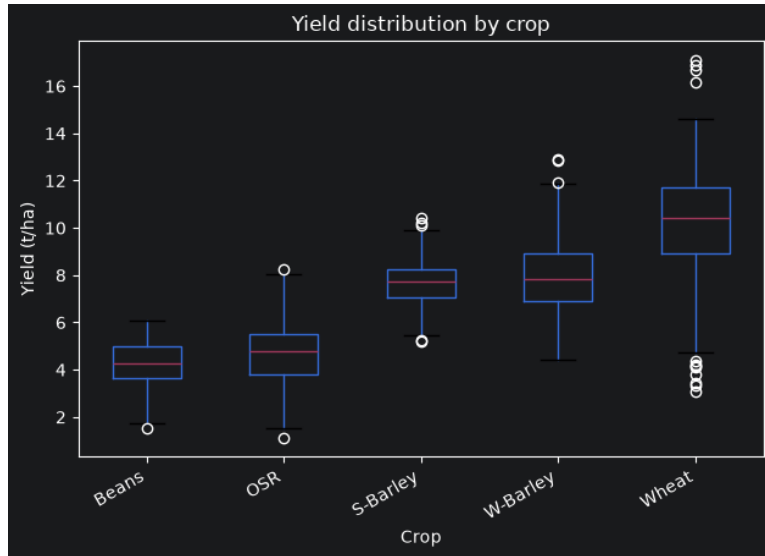
Yield + satellite (936 rows): one row per field per year, 2015 to 2017.

Climate: daily precipitation and temperature per grid square. Two real quirks handled here: the raw files are wide (one column per day), so it averages across days first; and every field sharing a 1km grid square gets an identical row, so it deduplicates on `grid_ID` before joining, otherwise a many-to-many join inflates the numbers.

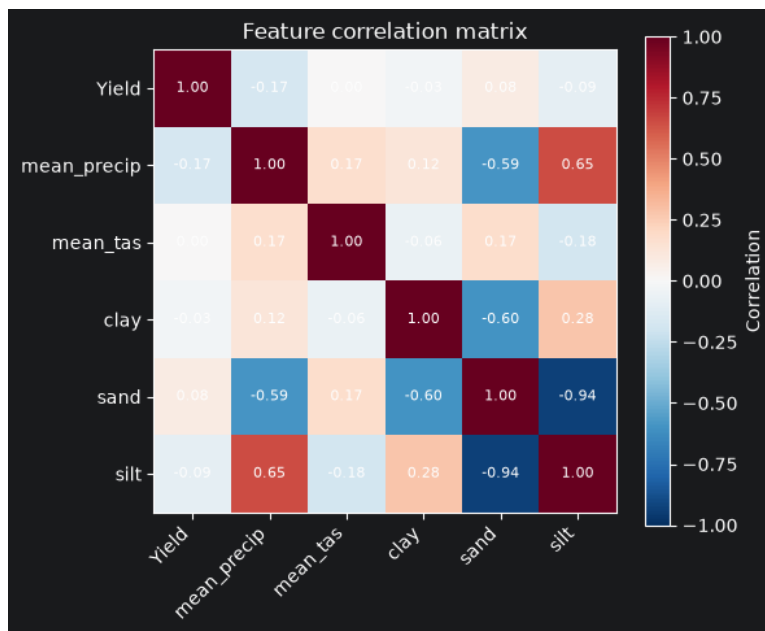
Soil: static per field. Uses clay/sand/silt directly rather than the text column, which looks like a USDA soil class but is actually a continuous numeric code, not a category.

Exploratory context

Before modelling, two charts set context for the audience.



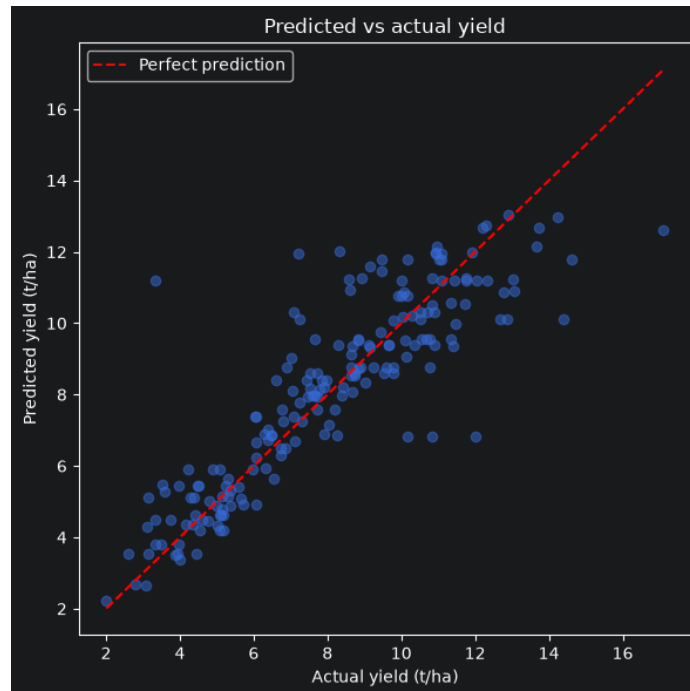
Wheat has both the highest yield and the widest spread of any crop, an intuition for what "explains yield" even means here.



Correlation matrix across Yield and all candidate features. Nothing pathologically redundant among the numeric features, so no multicollinearity concern going into the model.

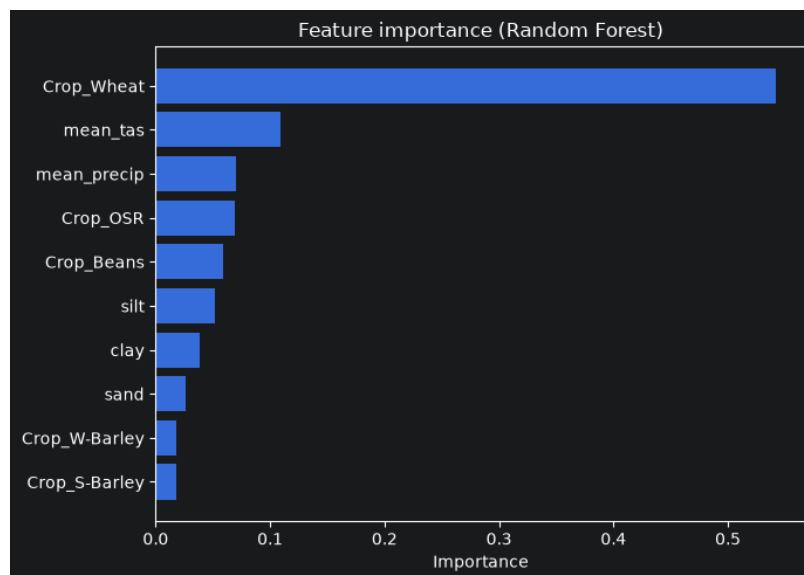
The model

A RandomForestRegressor predicting Yield from precipitation, temperature, clay/sand/silt, and crop type (one-hot encoded), on an 80/20 train/test split. Result: $R^2 = 0.760$, MAE around 0.99 t/ha. The model explains about 76% of yield variance, and is typically within a tonne per hectare on unseen fields.



Predicted vs. actual yield on the held-out test set. Tight along the diagonal in the middle of the range, with more scatter at the low and high extremes, as expected for tree-based models.

Feature importance



Feature importance from the random forest.

Crop_Wheat alone accounts for 0.54 of total feature importance, more than every climate and soil variable combined. The model is mostly learning "what yield range does this crop typically fall in," not "which climate/soil conditions drive yield." That isn't a flaw, it's an honest and expected finding, but it's worth stating explicitly rather than letting the chart speak for itself. It also sets up a natural next step: a per-crop model, or partial dependence plots controlling for crop, would show whether climate/soil actually matter within a crop type, arguably a more interesting research question than the current global ranking answers. Temperature (0.109) also edges out precipitation (0.070) as the next most useful signal.

Honest limitations

The satellite band values themselves (VV/VH/Ratio), arguably the actual point of the CYCLeSS dataset, aren't used as features yet, only climate/soil/crop.

The Scivision loader is compatible in format, not actually submitted to the Scivision catalog.